

Weather Forecast Application using React

Problem Statement

People often need quick access to weather information to plan their daily activities. This project demonstrates how to build a real-world React application that consumes data from an external API and displays weather information in a user-friendly interface.

Objectives

- Build a weather application using React.
- Learn API integration using Fetch API.
- Manage state using React Hooks.
- Handle loading and error states.
- Create a responsive UI.

Features

- Search weather by city name.
- Display temperature, humidity, and wind speed.
- Show weather icons.
- Loading and error handling.
- Responsive design.

Technology Stack

Frontend: React.js, CSS3
API: OpenWeatherMap API
Tools: VS Code, Git, GitHub

Project Flow

User enters city name → React captures input → API request sent → Weather data received → Data stored in state → Weather displayed

Folder Structure

```
src/  
■■■ App.jsx  
■■■ App.css  
■■■ components/  
■■■ assets/
```

Learning Outcomes

React Components, useState Hook, API Integration, Async/Await, Conditional Rendering, Responsive Design.

Future Enhancements

5-day forecast, current location weather, dark mode, charts, search history.